

AI-Powered QA/QC Document Lifecycle Automation System

A two-phase intelligent document management system that automates the complete QA/QC document lifecycle — from form submission and PDF generation to electronic signatures, automated filing, AI-assisted classification, and audit-ready archiving. Engineers fill in a form. The system handles everything else.

What it does	Automates QA/QC document creation, signing, submission, WhatsApp notifications, AI classification, and archiving
Tools used	n8n · Google Forms · Google Sheets · iLovePDF API · Gmail · Google Drive · Google Gemini AI · Meta (WhatsApp Business API)
Who benefits	QA/QC Engineers, Document Controllers, and Audit Teams in construction and engineering
Key features	Auto PDF generation · Electronic signatures · AI document classification · Audit-ready filing · WhatsApp notifications
Built by	Sydney Rey De Leon · AI & Automation Specialist Registered Civil Engineer

What Is This Project?

In construction and engineering, QA/QC documentation is not optional — it is a legal and professional requirement. Every inspection, test, and approval needs to be documented, signed, submitted to the right department, and stored in a way that can be retrieved and audited at any point. When that process is done manually, it is slow, inconsistent, and full of risk. Forms get filled out wrong. Documents sit waiting for signatures. Files end up in the wrong folders. Audit trails fall apart.

This system was built to solve that entirely. It is a two-phase automated workflow that covers the complete document lifecycle. **Phase 1** handles everything from the moment an engineer fills out a form — generating the PDF, inserting the electronic signature, merging attachments, and submitting the final document to the concerned department automatically. **Phase 2** picks up from there — monitoring incoming approved documents, using **Google Gemini AI** to classify and extract the content, creating the correct folder structure, and filing everything in an audit-ready archive. **From creation to archiving, no manual document handling is required.**

Built by a Registered Civil Engineer: QA/QC documentation is a daily reality in construction work. This system was not designed in theory — it was built with a direct understanding of how document control actually works on engineering projects, and what makes it break down when done manually.

2	100%	AI	0
Phases	Auto-Generated	Classification	Manual Filing
Document generation and intelligent archiving	PDFs created from form data, no manual typing	Gemini AI reads and sorts every document	Folder creation and archiving fully automated

The Two Phases at a Glance

The system is divided into two connected phases. Phase 1 focuses on producing and submitting the document. Phase 2 focuses on receiving, classifying, and permanently filing it. Together they cover the full document lifecycle:

Phase 1 — Document Generation & Submission

- Automatic PDF generation from user forms
- Electronic signature integration
- Attachment merging using the iLovePDF REST API
- Automated document packaging
- Automatic submission to the concerned department
- WhatsApp notification via Meta Business API

Phase 2 — Document Control & Final Filing

- Monitoring approved and signed documents
- Automated attachment processing
- AI-assisted document classification and extraction
- Automatic folder creation and file organisation
- Structured archival of records for traceability
- Audit-ready document filing system
- WhatsApp notification on successful filing

Phase 1 begins the moment an engineer fills out a QA/QC form. Everything that follows — generating the document, applying the signature, merging files, and sending the final package — happens automatically, without anyone touching a single file manually. The same process runs for each document type: MIR, IRI, SDA, NCR, and others.

1

An engineer fills out the QA/QC form

The process begins with a Google Form — a straightforward digital form that captures all the required QA/QC information: inspection details, date, location, responsible parties, and any relevant observations. Engineers fill this out on-site or at their desk. That submission is the only manual step in the entire process.

2

The system reads the form data and prepares the document

The moment the form is submitted, n8n picks it up and reads the response data from Google Sheets, where Google Forms automatically stores all submissions. The relevant fields are extracted and formatted, ready to be built into the official QA/QC document.

3

A professional PDF is generated automatically

Using the structured form data, the system generates a properly formatted PDF document. This is not a simple text export — it is a fully laid-out document that matches the official QA/QC template, with all fields populated correctly from the form data. No manual typing, no copy-pasting, no formatting.

4

The electronic signature is inserted into the document

The appropriate electronic signature is automatically applied to the PDF. This step uses an HTTP request to attach the signature in the correct position on the document, meeting the formal signing requirement without anyone needing to open the file or use a separate signing tool.

5

The signed PDF is converted and merged with attachments

The signed document is processed using the **iLovePDF REST API** — a professional PDF processing service. Any supporting attachments are merged with the main document to produce a single, complete file package. This ensures everything that belongs together arrives together.

6

The final document is uploaded and labelled

The merged, signed PDF is uploaded to the designated storage location and labelled according to the document type — MIR, IRI, SDA, NCR, or others. The file naming and labelling happen automatically based on the form data, so there is no risk of misfiling or inconsistent naming.

7

The complete document package is submitted automatically

A notification or email is sent to the concerned department with the final document attached and ready for review. The engineer does not need to manually email anyone, attach files, or follow up. The submission happens the moment the document is ready — instantly, every time.

What this replaces: Manually filling out a document template, printing it for signature, scanning it back, attaching supporting files, and emailing the package to the right department — repeated for every single QA/QC document across multiple document types. Phase 1 compresses that entire process into one form submission.

Phase 2 begins when the concerned department sends back an approved, signed document via email. From that point, the system takes over — processing the attachment, using AI to understand what the document is, creating the right folder structure, and filing everything in a permanent, traceable archive that is ready for audit at any time.

8

The system monitors for incoming approved documents

An email trigger watches the designated inbox for incoming approved documents from the concerned department. The moment an email with a relevant attachment arrives, the automation wakes up and begins processing. No one needs to manually check the inbox or forward the email to anyone.

9

The email thread is verified and the attachment is extracted

The system checks that the email belongs to the correct thread and that it contains an attachment. If an attachment is found, it is extracted automatically and passed forward for processing. If the email is not relevant, the system ignores it and continues monitoring.

10

The attachment is downloaded and prepared for AI analysis

The attached document is downloaded and converted into a format that can be analysed. PDF documents are rasterised into images so that the AI model can visually read the content of the document — including any stamps, signatures, handwritten notes, or form fields that a text-only reader might miss.

11

Google Gemini AI reads and classifies the document

This is where the intelligence of Phase 2 lives. **Google Gemini AI** is given the document images and instructed to read through them. It identifies what type of document it is — MIR, IRI, SDA, NCR, or others — extracts the key metadata such as project name, date, document number, and inspection details, and returns a structured classification that the system can act on. No human needs to read the document to decide where it should be filed.

12

The correct folder structure is created automatically

Based on the AI's classification output, the system creates or locates the correct folder in Google Drive. If the folder for this document type, project, or date does not yet exist, it is created automatically with the right naming convention. The folder structure is consistent across every project and every document — no matter who submitted it or when.

13

The document is filed and renamed correctly

The approved, signed document is uploaded into its designated folder with a standardised file name derived from the AI-extracted metadata. The naming is consistent, descriptive, and traceable — so anyone looking for a specific document later can find it without having to open dozens of files.

14

The record is archived and the team is notified

Once filed, the document is formally recorded in the archive log. A notification is sent to confirm the filing is complete — giving the team confidence that the document has been received, classified, and stored correctly. The archive is structured to support a full audit trail: who created it, when it was approved, and exactly where it is stored.

Audit readiness by design: Every document in the archive is filed consistently, named correctly, and linked to a traceable record. If an auditor or regulator requests any document — from any project, any date, any document type — it can be located immediately. No searching through inboxes, no asking colleagues where they saved it.

The Tools Working Together

Eight tools work together across both phases. Each one handles a specific part of the document lifecycle:

n

n8n — The automation engine

n8n is the open-source workflow platform that powers both phases. It connects every tool, runs the document generation logic, manages the email monitoring trigger, and orchestrates the AI classification and filing workflow from start to finish.

G

Google Forms & Sheets — The data entry and storage layer

Engineers submit QA/QC data through Google Forms. All responses are stored automatically in Google Sheets, which n8n reads to extract the form data needed to generate each document.

i

iLovePDF REST API — The PDF processing engine

iLovePDF is a professional PDF API used to merge the signed document with its supporting attachments, producing a single, complete file package ready for submission. It handles all PDF manipulation — compression, merging, and conversion — reliably and at scale.

G

Gmail — The communication and monitoring layer

Gmail serves two roles: it sends the completed document package to the concerned department in Phase 1, and it monitors the inbox for incoming approved documents in Phase 2. The email trigger in Phase 2 is the starting point for the entire archiving workflow.

M

Meta (WhatsApp Business API) — The instant mobile notification layer

The Meta WhatsApp Business API is used to send real-time notifications directly to team members' phones via WhatsApp. A message is triggered at two key moments: when a new document is submitted in Phase 1, and when an approved document is successfully filed in Phase 2. Because WhatsApp is already on everyone's phone, no one misses an update — even when they are away from their desk or on site.

Google Drive — The document storage and archive

G

All generated documents in Phase 1 and all approved documents in Phase 2 are stored in Google Drive. The folder structure is created automatically based on document type, project, and date — ensuring a consistent, navigable archive that scales across projects.

Google Gemini AI — The intelligent document classifier

G

Gemini AI reads the approved documents in Phase 2 and identifies what they are, extracting the metadata needed for correct filing. It removes the need for a human document controller to manually read and sort every incoming file.

HTTP / Webhook Requests — The signature and API integration layer

H

Custom HTTP requests handle the electronic signature insertion in Phase 1, connecting to the signature service with the correct parameters to apply the right signature to the right position on each document type.

Who Benefits from This System?

This system was built with the realities of engineering project work in mind. Three groups benefit directly:

QA/QC Engineers

Filling out one form is now the entirety of your document submission responsibility. The PDF is generated, signed, packaged, and submitted automatically. No printing, no scanning, no emailing — just the form.

Document Controllers

Incoming approved documents are classified, named, and filed automatically by AI. The folder structure is always consistent. The archive is always up to date. No manual sorting, no misfiled documents, no time spent organising a shared drive.

Audit & Compliance Teams

Every document in the archive has a clear trail — creation date, approval date, document type, project, and exact file location. When an audit happens, the records are already exactly where they need to be.

Why This Matters

Engineers spend zero time on document admin

- ✓ Filling out a form is the only thing an engineer needs to do. The entire document production and submission process runs automatically from that single action.

Every document is consistent and correctly formatted

- ✓ Because documents are generated from a template using form data, there are no formatting errors, no missing fields, and no inconsistencies between documents submitted by different engineers.

AI removes the need for manual document sorting

- ✓ Google Gemini reads and classifies every incoming approved document, so no one has to manually decide where it belongs or what to name it. The right filing decision is made automatically, every time.

The archive is always audit-ready

- ✓ Documents are filed consistently from day one. If an auditor requests any record at any time, it is exactly where it should be — correctly named, correctly categorised, and fully traceable.

The system works for multiple document types simultaneously

- ✓ MIR, IRI, SDA, NCR, and other document types all run through the same pipeline in parallel. The system handles each type correctly based on the form data and AI classification — no separate workflows needed.

It scales with the project without adding admin overhead

- ✓ Whether a project generates 50 documents or 5,000, the system processes every single one the same way. Growing project volume does not mean growing administrative burden.